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ORIGINAL ARTICLE

Reproducibility in measuring physical activity in children and adolescents with an acquired brain injury

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Abstract

Aim: To examine the reproducibility in measurement of physical activity performance using the ActiGraph® GT3X+ accelerometer in children aged 8–16 years with Acquired Brain Injury (ABI). **Methods:** Reproducibility of standardized tasks: Thirty-two children with ABI (12 years 1 month, SD = 2 years 4 months; 20 males; Gross Motor Function Classification System I = 17, II = 15) performed the following activities on 2 consecutive days while wearing an accelerometer and a heart rate monitor: quiet sitting, slow walking (SW), moderate walking (MW), fast walking (FW) and rapid stepping on/off a block (STEP). Intra-class correlation coefficients (ICC) were calculated. Performance variability: Fifty-one participants (12 years 1 month, SD = 2 years 5 months; 27 males; GMFCS I = 26, II = 25) wore an accelerometer for 4 days in the community and reliability coefficients were calculated using standardized 12-hour time spent in moderate-to-vigorous physical activity (MVPA).

Results: Test–re-test reproducibility was excellent for all activities (SW, ICC = 0.90; MW, ICC = 0.83; FW, ICC = 0.91; STEP, ICC = 0.89). Three days of monitoring produced excellent variability estimates of MVPA ($R = 0.78$).

Conclusion: Therapists can confidently use accelerometry as a reproducible measure of physical activity under standardized walking and stepping conditions, as well as in the community for children with ABI.

Keywords

Physical activity, children, adolescents, acquired brain injury, accelerometer

History

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Introduction

Children with an Acquired Brain Injury (ABI) may present with physical, cognitive and psycho-social deficits impacting their involvement in regular physical activity [1,2]. Children with an ABI are also significantly less physically active than typically-developing children (TDC). The prevalence of obesity for children with an ABI is higher than TDC (19% compared to 5%, respectively) [3–5]. In addition, these children may also present with attention and concentration difficulties which further impact their ability to consistently participate in regular leisure and sporting activities with typically-developing peers [6,7]. The benefits of physical activity for all children and adolescents include increased physical fitness, reduced body fat, more favourable cardiovascular and metabolic disease risk profiles, enhanced bone health and reduced symptoms of anxiety and depression [8]. It is, therefore, important to implement effective physical activity interventions using valid and reliable objective outcome measures to optimize the health and physical functioning of children with an ABI.

Habitual physical activity or physical activity performance refers to what an individual does in their natural environment

[9]. Physical activity performance in children and adolescents can be measured using a wide range of subjective and objective methods [10]. Subjective methods include self-report questionnaires, activity logs and diaries and objective methods include direct observation, doubly labelled water, heart rate monitors, accelerometers and pedometers [10]. Self-reported methods are subject to social desirability and recall bias, while objective measures such as indirect calorimetry and direction observation are not feasible in free-living physical activity studies [10,11]. Accelerometers have, therefore, become the most widely used objective measure of physical activity in children [12,13].

The ActiGraph® accelerometer is a valid and reliable measure of ambulatory physical activity performance in children and adolescents with neurological disorders [14–16]. The ActiGraph® 7164 accelerometer has concurrent validity with oxygen consumption ($\rho = 0.70$ – 0.85) and excellent inter-instrument reliability (intra-class correlation coefficients [ICC] = 0.94 – 0.99) during standardized activity trials (reflecting daily routines) in children with cerebral palsy (CP) [14]. The ActiGraph® GT3X+ accelerometer also has good-to-excellent reliability with stepping tasks (ICC = 0.66), light and moderate walking (ICC = 0.8) and vigorous walking (ICC = 0.7) under laboratory conditions in independently ambulant children and adolescents (aged 8–16 years) with unilateral CP [15]. It can also differentiate between varying

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intensities of walking (slow, comfortable and brisk paced) under standardized conditions in children and adolescents with CP using accelerometer-based estimates of Metabolic Equivalents (METs) [16]. The use of accelerometers to reliably measure physical activity during standardized walking and stepping tasks in children and adolescents with an ABI has not been reported. It is important to determine reliability of accelerometers in order to accurately interpret the significance of physical activity interventions.

It has been recommended in TDC that between 4–5 days of monitoring are required to reliably estimate daily physical activity levels (Spearman-Brown analysis of 0.80) [17]. In children and adolescents with CP, 3 days of monitoring in a free-living situation have been recommended to produce reliable estimates of moderate-to-vigorous physical activity (MVPA) [15]. Both weekdays and weekends should be included in the monitoring period due to marked differences in physical activity behaviour [18]. The reliability of ActiGraph® accelerometers during free-living situations as well as the number of days of monitoring needed have not been determined in children and adolescents with an ABI. It is important to determine how many and what type (weekday or weekend) of days are required to determine physical activity levels (more specifically, time spent in MVPA) in this population for future use in clinical and research settings.

The primary aim of this study was to investigate the reproducibility (ICC) of recording physical activity using the ActiGraph® GT3X+ tri-axial accelerometer during repeated controlled activity intensities under standardized laboratory conditions in independently ambulant children and adolescents aged 8–16 years with an ABI. Second, the study aimed to determine the variability in MVPA and number of days of monitoring required to measure physical activity performance in the community with sub-group analyses investigating differences between age groups (children aged < 13 years compared to adolescents aged ≥ 13 years). Third, this study explored differences in daily time spent in MVPA compared to day type (weekday vs weekend), age, gender and Gross Motor Function Classification System (GMFCS) level.

It was hypothesized that the ActiGraph® accelerometer would have excellent reproducibility in children and adolescents with an ABI. To reliably estimate physical activity levels, it was hypothesized that 3 days of monitoring would be needed and children would demonstrate more variability than adolescents. Finally, daily MVPA levels would be significantly higher for weekdays, children, males and those equivalent to GMFCS I.

Methods

Participants and recruitment

Children and adolescent participants were included in this study if they were at least 12 months post an ABI which was acquired ≥ 28 days post-birth. They were also required to be: (i) independently ambulant and equivalent to GMFCS equivalent level I (walks without limitations) or II (walks with limitations); (ii) aged 8–16 years; and (iii) co-operative and have sufficient cognitive understanding, visual and verbal abilities to perform the required tasks. Children were excluded if they had a degenerative or metabolic condition, unstable epilepsy (i.e. frequent, daily seizures not controlled or not yet managed by medication) or had undergone any

surgical or medical intervention (e.g. chemotherapy or radiotherapy) in the previous 2 or 6 months, respectively. This study was embedded in a larger randomized controlled trial (Mitii™ ABI) [19]. For the collection of repeated measures during standardized tasks, a sample of 32 children were recruited via consecutive series sampling from the larger Mitii™ ABI study and assessed during baseline assessments over 2 days [19]. Written and informed consent was obtained from parents/guardians and assent from participants if they were 12 years or older. Full ethical approval was obtained from the Medical Ethics Committee of The University of Queensland (2013000212) and The Royal Children's Hospital Brisbane, Australia (HREC/12/QRCH/222).

Sample size

To detect an ICC of $\rho > 0.6$ (against a null hypothesis of $\rho_0 = 0.2$) with significance set at 0.05 and power at 0.80, it was determined that 27 participants were required for the reproducibility of standardized tasks under controlled conditions [20]. In total, 32 children with an ABI were recruited. Four day accelerometer data collected during baseline assessment for all participants of the Mitii™ ABI randomized controlled trial were used to determine the number of day requirements and daily MVPA [19].

Instrumentation

To evaluate reproducibility between repeated measurements under controlled laboratory conditions, participants were required to wear an accelerometer and a heart rate (HR) monitor concurrently. For measurement of physical activity performance variability between days in the community, participants only wore an accelerometer.

Accelerometer

A tri-axial accelerometer (ActiGraph® GT3X+; Pensacola, FL) sampled and stored acceleration data for three axes at 100 Hz. The raw acceleration data were processed into counts per 5-seconds by the ActiLife™v6 (ActiGraph® Corporation) software. Before commencement of each trial, the accelerometer was initialized and attached to an elastic belt which was worn by the participant around the waist. The accelerometer was positioned on the mid-axilla line at the level of the anterior superior iliac spine on the side of the body with the least neurological impairment [16]. Each participant wore the same accelerometer unit over consecutive assessment days; however, accelerometer units varied between participants.

Heart rate monitor

Heart rate was recorded in 5-second epochs using a Polar RS400® monitor (Polar Electro Oy, Kempele, Finland). The monitor was positioned according to the manufacturer's instructions with the chest strap moistened and placed around the chest in line with the middle of the sternum.

Reproducibility of standardized tasks

Procedure

The testing protocol comprised an initial 5-minute seated rest period (while watching a movie on a tablet device) to establish resting heart rate and accelerometry values, followed by four standardized tasks: (1) 5 minutes of slow walking (SW, 0.33 m s^{-1}); (2) 5 minutes of moderate paced walking (MW, 0.67 m s^{-1}); (3) 6 minutes of fast paced walking (FW) alternatively called the 6-Minute Walk Test (6MWT). Participants were instructed to ‘walk as far as possible without running in 6 minutes’ and; (4) 3 minutes of rapid stepping on/off a 20 centimetre block (STEP). Walking tasks were performed on a 10 metre carpet track with cones demarcating the turning points. During the SW and MW trials, study personnel walked alongside the participant to monitor comfort and assist them to maintain a constant speed. When variations in walking speed occurred, this was noted and study personnel asked the participant to increase or decrease walking speed accordingly. For the 6MWT, participants were given verbal encouragement every minute as well as distance covered (in laps) and time remaining. Distance was measured to the nearest 1 metre. Following the walking trials, participants completed a stepping task for 3 minutes using a 20-centimetre high aerobic step block, being instructed ‘to step on and off the step block as quickly as possible for 3 minutes’. Between each task and at the conclusion of the protocol, participants rested for 5 minutes in a seated position watching a movie on a tablet device. The current procedure was based on that described in Clanchy et al. [16] and Mitchell et al. [15]. To assess reproducibility of the accelerometer, the testing protocol was repeated on 2 consecutive days.

Data reduction

Raw HR and accelerometer data were synchronized, compiled and analysed using customized MATLAB (MathWorks, USA, 2014a) scripts. Mean HR and vector magnitude of the middle 2 minutes of each rest and activity intensity period were calculated.

The standardized tasks were classified into METs using the scheme: sedentary activity < 1.5 METs; light ≥ 1.5 METs; moderate ≥ 3.0 METs and; vigorous activity ≥ 5 METs (Table I) [21,22].

Statistical analysis

Continuous data were summarized as means and standard deviations (SD) and activity counts were presented in counts per 5-second epoch unless stated otherwise. Reliability between repeated measurements was calculated using ICC between days for each task (two-factor mixed effects agreement model) from day 1 and day 2, within each activity task (sedentary, SW, MW, FW and STEP). Intra-class correlation coefficients greater than 0.75 were interpreted as excellent, between 0.4–0.75 as acceptable, and less than 0.4, poor [23]. The correlation between activity counts obtained by the accelerometer and increase in HR for each task was calculated using Pearson’s correlation coefficients (ρ).

Performance variability

Procedure

Children wore the accelerometer for all waking hours for 4 days (consisting of 2 week days and two weekends) with the exception during water-based activities. During this time, parents recorded the wear time of the accelerometer and activity patterns, on an activity diary provided.

Data reduction

Data from the accelerometer were downloaded using the ActiLife™ v6 software into 60 second epochs. Participant accelerometer data were included in further analysis if daily wear time was greater than or equal to 8 hours or greater than 7 hours and following review of the activity diary, the accelerometer had been removed to perform water-based activities or sports which prohibited children wearing an accelerometer. Customized MATLAB scripts were used to process the 4-day accelerometer data. Where activity count values of zero occurred for > 20 minutes, this was set as non-wear time and excluded from the analysis. Vector magnitude activity counts were converted to activity intensity using child ABI-specific cut-points (originally derived in 15 second epochs and converted to 60 second epochs) for sedentary (≤ 400), light (> 400) and MVPA (≥ 1876) [24] and standardized into 12-hour wear time to account for differences in wear time between participants.

Statistical analysis

Independent two-sided *t*-tests were calculated to investigate differences in age, gender, GMFCS and day type (weekday/weekend) using standardized 12-hour time spent in MVPA for each day per participant. Variability was calculated using between-day intra-class reliability coefficients (*R*) and 95% confidence intervals (CI) using repeated measures analysis of variance (ANOVA). The required days of monitoring needed to achieve reliabilities of 0.70, 0.80 and 0.90, respectively, were calculated using the Spearman-Brown prophecy formula:

$$\begin{aligned} \text{Days of monitoring needed for desired reliability} \\ = (ICC_{\text{desired}} \div [1 - ICC_{\text{desired}}]) \\ \times (1 - ICC_{\text{estimated}} \div [1 - ICC_{\text{estimated}}]) \end{aligned}$$

where $ICC_{\text{estimated}}$ was 2-day reliability (formula multiplied by a factor of two) [10,15,25]. Sub-analyses were conducted to determine the difference in reliability estimates between children (aged 8–12 years) and adolescents (aged 13–16 years). Data were analysed using IBM SPSSv22 (IBM Corp.; Armonk, NY).

Results

Reproducibility of standardized tasks

Participant demographics and classification for the 32 participants in the reproducibility of standardized tasks are reported in Table II. Mean and SD for activity counts (per 5-second epoch), speed and HR across days are presented in Table III. One outlier was excluded from HR data during SW as the data point significantly

influenced results. Activity counts and HR were significantly correlated between days for all standardized tasks (Table III). Test–re-test reliability for each activity intensity task is presented in Table IV (SW, ICC = 0.90; MW, ICC = 0.83; FW, ICC = 0.91; STEP, ICC = 0.89).

Performance variability

In total, 60 children with an ABI were enrolled in the MitiTM ABI randomized controlled trial [19] and 58 completed baseline measurements. ActiGraph[®] recordings were complete and valid for 51 participants (88%). The demographics and classification for the 51 participants are reported in Table II. Between day intra-class reliability coefficients indicated that between 3–4 days of monitoring produced excellent estimates

of daily MVPA ($R = 0.78$ – 0.79 ; Table V). Two days of monitoring produced more variability ($R = 0.72$ [0.52–0.84]). Spearman-Brown analysis estimated that 2 days of monitoring would be necessary to achieve a reliability coefficient of 0.70, and 7 days to achieve a reliability coefficient of 0.90. Adolescents 13 years and older were less variable in their physical activity performance between days, with 2 days providing acceptable estimates of daily MVPA ($R = 0.81$ [0.49–0.93]). Younger children had more variable estimates of daily MVPA, with 3 days of monitoring required to produce acceptable estimates of daily MVPA ($R = 0.73$ [0.52–0.86]). These results demonstrated that 4 days of monitoring would be necessary to achieve a reliability coefficient of 0.70 in children and 2 days for adolescents (Spearman-Brown analysis).

Secondary analyses investigating differences between standardized (12 hour) time spent in MVPA by age, gender, GMFCS and day type are presented in Table VI. There were significant differences in time spent in MVPA between children and adolescents. There were no significant differences in time spent in MVPA between gender, GMFCS (I or II) or day type (weekday or weekend).

Discussion

This study demonstrated excellent test–re-test reproducibility with ICC of 0.83–0.91 using the ActiGraph[®] GT3X+ accelerometer for repeated measurements of standardized activities in independently ambulant children and adolescents aged

Table I. Standardized movement protocol and corresponding activity level.

Activity intensity level	MET level	Task	Time (minutes)	Lap speed (m s ⁻¹)
Sedentary	< 1.5	REST	5	N/A
Light	> 1/5–3	SW	5	0.33
Moderate	3–6	MW	5	0.67
Vigorous	> 6	FW	6	>0.67
		STEP	3	N/A

FW, fast walking; MET, metabolic equivalent; MW, moderate walking; N/A, not applicable; REST, seated rest; STEP, stepping on/off block; SW, slow walking.

Table II. Study participant demographics and classification for repeated measurements.

Variable	Reproducibility of standardized tasks ($n = 32$)	Performance variability ($n = 51$)
Age, mean (SD)	12 y 1 mo (2 y 4 mo)	12 y 1 mo (2 y 5 mo)
Age at injury, mean (SD)	5 y 9 mo (3 y 10 mo)	5 y 8 mo (3 y 7 mo)
Intellectual ability, FSIQ, mean (SD)	78.41 (14.97)	77.14 (16.03)
AROC impairment codes, n (%)		
Stroke	14 (43.75)	20 (39.21)
TBI	10 (31.25)	15 (29.42)
Non-TBI	8 (25)	16 (31.37)
GMFCS, n (%)		
Level I	17 (53.13)	26 (50.98)
Level II	15 (46.87)	25 (49.02)

AROC, Australasian Rehabilitation Outcomes Centre; FSIQ, full scale intelligence quotient; GMFCS, Gross Motor Function Classification System; mo, months; non-TBI, non-traumatic brain injury; SD, standard deviation; TBI, traumatic brain injury; y, years.

Table III. Correlations between days for ActiGraph[®] accelerometer vector magnitude activity counts and heart rate by standardized tasks.

Task	Test, mean (SD)			Re-test, mean (SD)			Correlation day 1 and 2 (ρ)	
	Activity counts (counts/5 s)	Speed (m min ⁻¹)	HR (bpm)	Activity counts (counts/5 s)	Speed (m min ⁻¹)	HR (bpm)	Activity counts (counts/5 s)	HR
REST	0.35 (0.85)	N/A	91.3 (21.87)	0.52 (1.16)	N/A	91.30 (21.87)	N/A	N/A
SW	174.95 (39.34)	20	105.12 (15.16)	181.40 (47.65)	20	106.19 (14.43)	0.83	0.86
MW	261.18 (35.68)	40	110.17 (14.62)	262.64 (47.08)	40	110.20 (14.23)	0.74	0.88
FW	463.54 (92.84)	68.33 (10.91)	141.26 (21.77)	481 (96.25)	69.45 (11.33)	142.50 (22.0)	0.84	0.81
STEP	455.42 (156.04)	N/A	146.15 (18.40)	509.31 (183.95)	N/A	150.75 (21.80)	0.81	0.83

All correlations were significant to $p < 0.05$.

Bpm, beats per minute; counts/5 s, vector magnitude activity counts per 5 seconds; FW, fast walking; HR, heart rate; m min⁻¹, metres per minute; MW, moderate walking; N/A, not applicable; REST, seated rest; SD, standard deviation; STEP, stepping on/off block; SW, slow walking.

Table IV. Intra-class correlation coefficients between repeated measures for ActiGraph® accelerometer activity counts per minute.

Task	Vector magnitude activity counts	
	ICC	95% CI
SW	0.90	0.79–0.95
MW	0.83	0.66–0.92
FW	0.91	0.82–0.96
STEP	0.89	0.77–0.95

FW, fast walking; ICC, intra-class correlation coefficients; MW, moderate walking; STEP, stepping on/off block; SW, slow walking; 95% CI, 95% confidence interval.

MVPA ($R = 0.72$ – 0.79) for this study is comparable to literature in children with CP and TDC [15,26]. Four days of monitoring produced reliable coefficient estimates of 0.79, which is comparable to both children and adolescents with CP and TDC (0.79 and 0.75–0.78, respectively) [15,17]. Similar to children with CP [15], children with an ABI demonstrated larger variation in reliability estimates compared to adolescents with an ABI (children: 0.57–0.70; adolescents: 0.81–0.79). Children with an ABI required 3–4 days of monitoring, while adolescents only needed 2 days to achieve reliable estimates of daily MVPA levels. These results may be explained by a wider range of natural daily

Table V. Performance variability in moderate-to-vigorous physical activity based on days of monitoring in children and adolescents with an acquired brain injury.

	<i>n</i>	Variability, Reliability coefficient (95% CI)			Days to achieve reliability of		
		2 day	3 day	4 day	0.7	0.8	0.9
Total	51	0.72 (0.52–0.84)	0.78 (0.66–0.87)	0.79 (0.67–0.87)	1.81	3.11	7
Children	33	0.57 (0.13–0.79)	0.73 (0.52–0.86)	0.70 (0.48–0.84)	3.52	6.04	13.58
Adolescents	18	0.81 (0.49–0.93)	0.76 (0.48–0.90)	0.79 (0.56–0.91)	1.09	1.88	4.22

Adolescent, ≥ 13 years; child, < 13 years.

Table VI. Standardized 12-hour time spent in moderate-to-vigorous physical activity in independently ambulant children with an acquired brain injury and by age, gender, functional ability and day type.

	Total	Age			Gender			GMFCS level			Day Type		
		Child	Adolescent	P-value	Male	Female	<i>p</i> -value	I	II	<i>p</i> -value	Weekday	Weekend	<i>p</i> -value
Number (<i>n</i>)	51	33	18		27	24		26	25		135	64	
MVPA (hours)		2.11 (1.01)	1.35 (0.79)	< 0.01	1.93 (1.10)	1.74 (0.89)	0.19	1.85 (1.14)	1.83 (0.85)	0.84	1.77 (0.91)	1.98 (1.19)	0.17

Data are mean (SD).

Adolescent, ≥ 13 years; child, < 13 years; GMFCS, Gross Motor Function Classification System; MVPA, mean daily time spent in moderate-to-vigorous physical activity with wear time standardized to 12 hours.

8–16 years with an ABI. Accelerometers were accurately able to measure movement during standardized tasks with controlled (SW and MW) speeds and tasks with self-selected pacing (FW and STEP) consistently across repeated test days. The high level of reproducibility in measurement of physical activity performance using accelerometers is consistent with previous literature in school-aged children with CP [14,15]. Reproducibility of equivalent standardized tasks using the same type of accelerometer in children with CP produced ICC values ranging between 0.66–0.80 [15]. Results from this study, therefore, suggest that the reproducibility of standardized tasks across days using the ActiGraph® GT3X+ accelerometer is more accurate in children with an ABI than in children with CP [15].

The secondary aim of this study was to determine the variability in MVPA and number of days of monitoring required to measure physical activity performance in the community for children with an ABI. Overall variability in

MVPA variation in children with an ABI [15]. Children are more likely to engage in diverse MVPA such as climbing and playing games in free-living situations [17]. Alternatively, the results may have reflected more consistent physical activity behaviour in adolescents with an ABI as they are more likely to engage in regular MVPA activities such as gym programmes and running [17].

Secondary analyses demonstrated that children with an ABI performed significantly more MVPA during a 12-hour standardized time frame than adolescents. Children participated in ~ 45 more minutes of daily MVPA compared to adolescents with an ABI. These results are consistent with previous literature suggesting daily MVPA decreases with increasing age in both TDC and children with CP [26,27]. As social and educational demands and expectations increase as children grow older, this may influence daily time spent in MVPA [28]. In comparison to TDC and children with CP, however, children with an ABI also

demonstrate social, behavioural and cognitive impairments which may further impact involvement in physical activity. It is unknown whether physical activity levels reduce more as children with an ABI grow older than in TDC or children with CP. Although levels of MVPA were similar between weekdays and weekends, wide confidence intervals (for both weekdays and weekends) indicate large variability between participants which may reflect differing levels of physical, cognitive and psychosocial deficits impacting participation in home, school and community settings [28,29]. In particular, children with an ABI are most restricted in structured community events and activities and social and play activities with peers either at school or in the community [28,30].

The Australian national physical activity guidelines recommend that children and adolescents participate in 60 minutes of daily MVPA for health benefits [31,32]. Based on new child and adolescent ABI-specific cut-points developed for detecting sedentary, light and MVPA in children and adolescents with an ABI [24], the results of this study suggest that children and adolescents with an ABI are meeting these recommendations with 120 and 80 minutes of daily MVPA, respectively. These new cut-points yielded a lower threshold for detecting MVPA than the previous published cut-points for TDC [33] and children with CP [16] and highlight the importance of age and participant specific cut-points. The lower threshold for detecting MVPA may reflect a higher energy cost of walking for children and adolescents with an ABI compared to TDC and children with CP. Children with an ABI compared to TDC had greater oxygen consumption during velocity or distance controlled physical activity tasks [34]. The greater oxygen consumption equated to 42% more energy per step during walking tasks for the children with ABI than TDC [34]. This evidence suggests that children with an ABI may not need to perform walking activities at the same intensity or speed as their peers to reach the recommended 60 minutes of MVPA during activities of daily living. Children with an ABI may reach 60 minutes of MVPA by performing simple tasks such as walking to and from school, between classes and around the home environment. In contrast, their typically-developing peers may need to walk briskly, cycle to and from school and participate in extra-curricular activities (e.g. soccer, dancing or netball) to reach the recommended 60 minutes of daily MVPA. Whether these tasks will produce the same mechanical load to promote muscle and bone development in children with an ABI has not been investigated. Clinicians should, therefore, consider activity intensity and impairments (attention, concentration, and fatigue difficulties) when prescribing physical activities for children with an ABI.

The results of the current study need to be considered in light of some potential limitations. Test-re-test reproducibility for the accelerometer in ambulant children and adolescents with an ABI was only explored during walking and stepping tasks under controlled conditions. Therefore, reliability of other tasks (such as running, cycling and jumping) and in individuals with lower functional classification (GMFCS equivalent III, IV or V) requires further study. Additionally, inequality between the number of children and adolescents

may have influenced results. The Spearman-Brown prophecy formula assumes reliability increases with increased days. As reliability did not increase in the same proportion (and in some cases decreased) as predicted, results should be interpreted with caution.

Conclusion

To the authors' knowledge this is the first study to investigate the test-re-test reproducibility of objectively measured physical activity performance in independently ambulant children and adolescents aged 8–16 years with an ABI. Measurement of physical activity using the ActiGraph® GT3X+ accelerometer is reliable under standardized walking and stepping conditions, as well as in a community environment. Clinicians and researchers can, therefore, accurately interpret the clinical significance of interventions aimed at improving physical activity performance using accelerometry for this population. The findings suggest that 3 days of monitoring would produce acceptable estimates of reliability for children and adolescents with an ABI.

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Declaration of interest

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